# Swiggy Food Delivery Data Analysis

*Delivery Time, Order Patterns, and Menu Item Demand Trends*
Tracking Best-Sellers, SLA Risk, Peak Ordering Windows, Zone Demand & Availability Risk


| | |
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| year | **2nd** |
| Report type | **Descriptive analytics (What happened?)** |


## Note on Data

Live Swiggy order data is not publicly accessible, so this report runs on a synthetic dataset built to mirror real food-delivery order patterns: 10 menu items across 6 cuisine categories, 5 delivery zones, over a 15-day window (Jul 01, 2026 – Jul 15, 2026), for 150 item-day records in total.

Each row represents one menu item's aggregated activity for one day at its dominant delivery zone — orders, average delivery time, revenue, customer rating, the time-of-day band with the

heaviest ordering, and remaining kitchen availability level. This is the same granularity a real food-delivery analytics pipeline would expose at the reporting layer: raw per-order events are rolled up before they ever reach a dashboard (see Section 3).

- one menu item (Chicken Biryani) generates disproportionate order volume, mirroring the Pareto pattern common on food-delivery apps' most-loved comfort-food lines.
- Weekend and evening demand surges layered on top of small category-level growth trends.
- A slow-building SLA risk signal at one delivery zone, where average delivery time climbs past the 30-minute target over the window — the kind of trend an operations dashboard should surface early.
- Uneven zone-level demand concentration, and two menu items deliberately depleting into Low Availability / Sold Out states.

## Objective

Build a descriptive analytics layer over daily Swiggy order data that answers five recurring, operational business questions — the kind a category manager, kitchen-partner operations lead, or delivery-ops team would otherwise have to ask an analyst for every morning:

**Q1** What are the best-selling menu items, and by how much do they lead the pack?

**Q2** How are delivery times trending, and which delivery zone is at risk of breaching the 30-minute SLA?

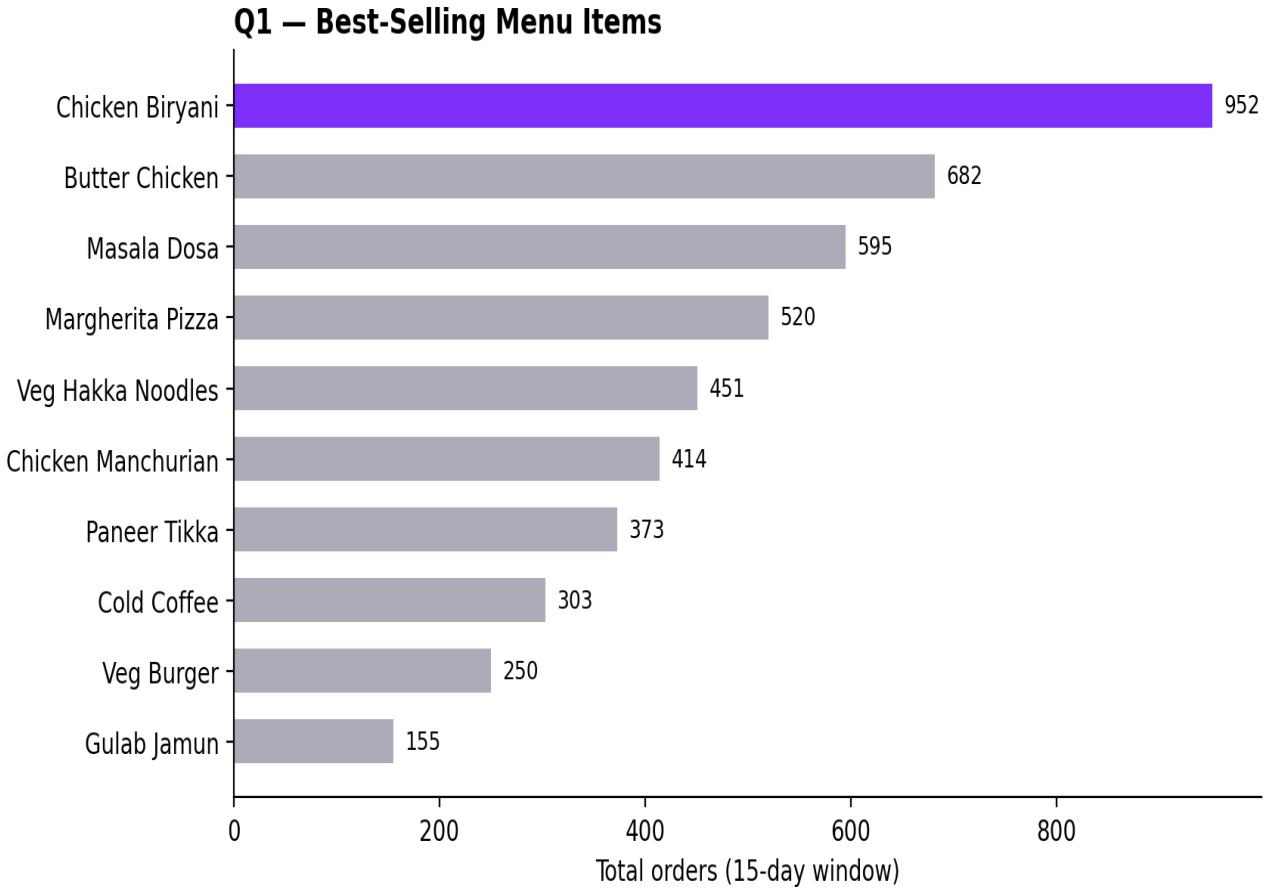
**Q3** When do customers order most — by time of day and day of week?

**Q4** Where is order demand concentrated across delivery zones, and what does that imply for fulfillment?

**Q5** Which menu items are at risk of going unavailable at the kitchens, given current order velocity?

## How This Would Work at Real Scale

# I. Best-Selling Menu Items




*Generated with pandas (aggregation) + matplotlib (visualization) directly from swiggy_order_data.csv.*

- **Chicken Biryani** led with **952 orders**, 39.6% ahead of the second-place item (**Butter Chicken**, 682 orders).
- The top menu item alone accounts for **20.3%** of all orders across 10 menu items — a classic *hero-dish* pattern where a small number of comfort-food listings drive a disproportionate share of order volume.

The analysis ranks the menu items based on their total number of orders over the 15-day period. Chicken Biryani was the best-selling item with 952 orders, followed by Butter Chicken with 682 orders. Chicken Biryani recorded 39.6% more orders than the second-best-selling item.

The top-selling item accounted for 20.3% of the total orders across all 10 menu items, showing a Hero-Dish effect, where a single item contributes a disproportionately large share of the total order volume.

This pattern is related to the Pareto Principle (80/20 Rule), where a small number of menu items can generate a large portion of overall demand. Therefore, high-demand items should receive greater attention in kitchen prep planning, ingredient stocking, and fulfillment operations.

Ranking logic here is orders, not revenue: a mid-priced comfort dish like biryani outselling a premium item in order count matters differently to kitchen-partner slotting (prep load, packing priority) than it does to finance (margin contribution) — a real dashboard should expose both cuts as separate views

With Chicken Biryani carrying 20.3% of total orders, it should get priority treatment on three fronts: it should be the first item checked in every kitchen-partner's prep readiness audit (a stockout here costs more than on any other item), it should sit at the front of the packing queue, and its delivery-time performance (Section 5) should be tracked separately from the zone average — a slow

hero dish delivery disproportionately damages the habitual on-time-delivery expectation that keeps customers ordering again.

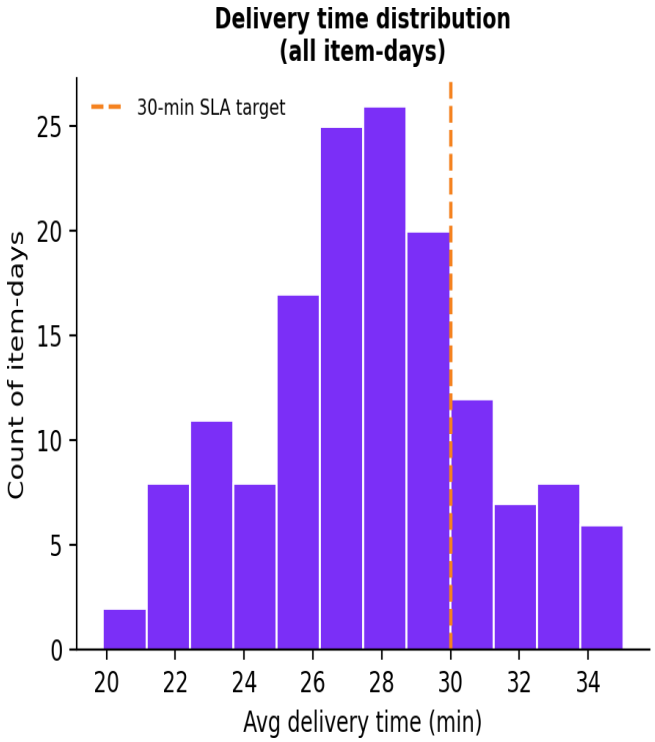
## KPI definitions used:

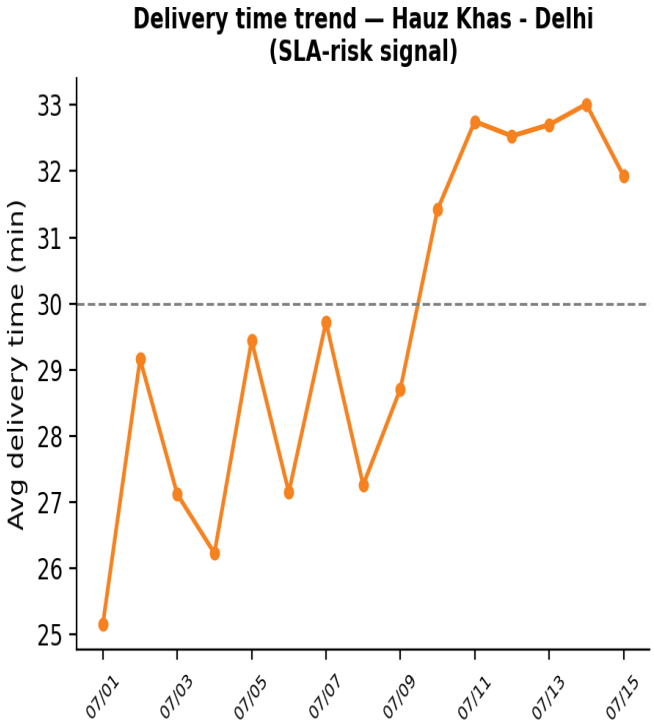
Total Orders = Sum of orders across all menu

items during the analysis period. Top Item Share =

Top-selling item orders ÷ total orders × 100.

# II. Delivery Time: Distribution & SLA Risk



The overall average delivery time was 27.6 minutes, compared with the 30-minute SLA target. 22.0% of item-days exceeded the 30-minute SLA, showing that delivery-time breaches were occurring frequently.

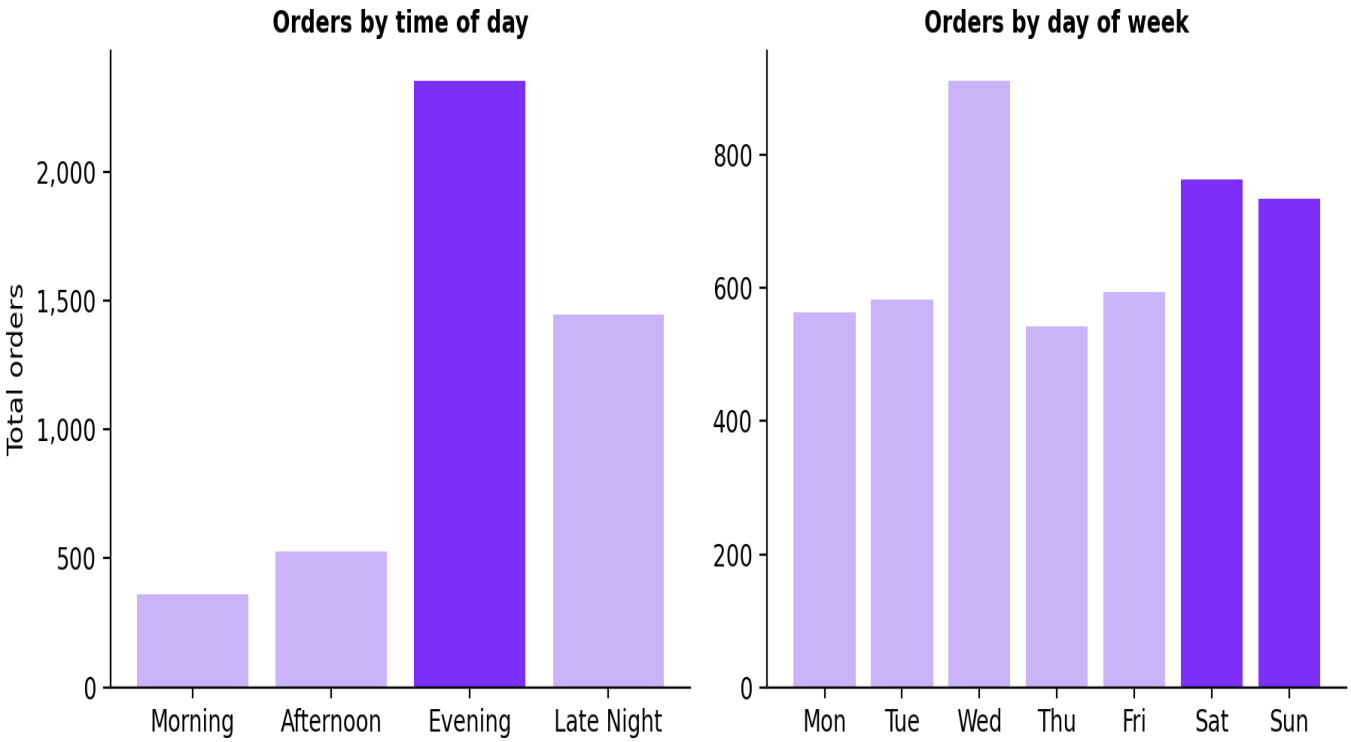
However, the average alone does not show whether delivery performance is improving or getting worse. Therefore, trend analysis is also used to observe how delivery time changes over the period.

Hauz Khas - Delhi was identified as the highest SLA-risk zone, where average delivery time increased from 25.2 minutes to 31.9 minutes during the 15-day period. This upward trend indicates a potential operational issue such as kitchen-partner capacity limitations or rider availability problems.

Hauz Khas - Delhi's delivery-time trend is the single clearest action item in this report: a climb from 25.2 to 31.9 minutes in 15 days, if it continued at the same rate, would push the zone well past the point where customers start noticing and churning — worth investigating (rider headcount in that zone, kitchen packing bottlenecks, or a demand spike outmatching current staffing) before the next weekly ops review rather than after.

In production this view is most useful segmented, not aggregated: zones trending upward in rolling 3-day delivery time, and zones combining high order volume with SLA breaches, are the two highest-priority slices — volume amplifies the customer-facing damage of a slow zone.

# Q3 — Order Patterns: Time of Day & Day of Week




*Generated with pandas (aggregation) + matplotlib (visualization) directly from swiggy_order_data.csv.*

- Evening (5pm-9pm) is the clear peak window, capturing 50.2% of all orders — consistent with food-delivery apps being used heavily for dinner-time orders and late-evening cravings.

Demand seasonality refers to recurring patterns in customer demand over different periods of time. In this analysis, order patterns are studied using two time scales: time of day and day of week.

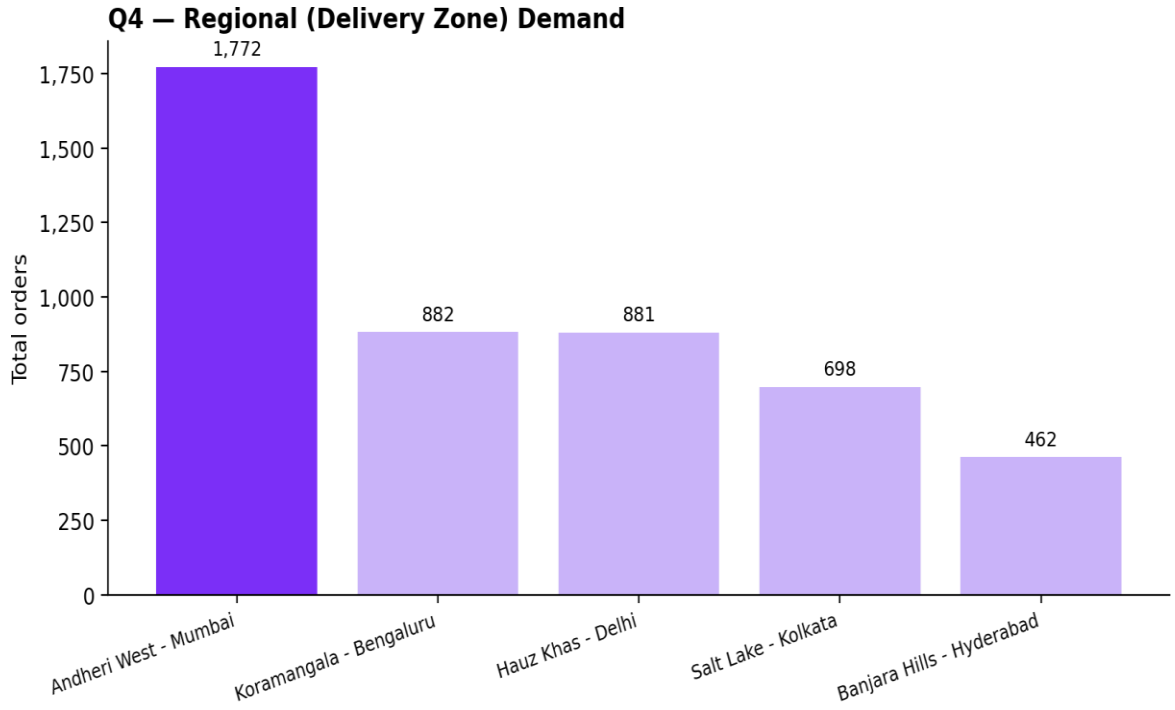
The analysis shows that evening (5 PM–9 PM) is the peak ordering period, accounting for 50.2% of all orders. This indicates that customer demand is highest during the evening, when customers commonly place orders for dinner, snacks, and late-night cravings.

The analysis also shows that weekend order volume runs 17.1% higher than the average weekday, reflecting more discretionary and social-occasion ordering (comfort food, desserts, beverages) on Saturdays  and Sundays compared to the routine weekday ordering pattern.

These patterns represent demand seasonality at multiple time scales. Time-of-day patterns are useful for rider shift-scheduling and kitchen-partner staff allocation, while day-of-week patterns help with ingredient pre-positioning and kitchen-partner staffing decisions made a day or more in advance.

With 50.2% of orders landing in the evening window, rider shift schedules and kitchen-partner packing rosters should be weighted toward that window rather than spread evenly across the day — the delivery-time risk in Section 5 is also most likely to materialize during this same peak, since capacity is stretched thinnest exactly when order volume is highest.

# Q4 — Regional (Delivery Zone) Demand




*Generated with pandas (aggregation) + matplotlib (visualization) directly from swiggy_order_data.csv.*

- **Andheri West - Mumbai** leads with 1,772 orders (37.7% of total demand), noticeably ahead of the other 4 zones tracked.
- Demand tapers off fairly smoothly from the top zone down — at real Swiggy-scale volumes across a full city network, this concentration is typically sharper still, with a handful of dense residential/office zones accounting for the large majority of orders and newer or lower-density zones long-tailed.

- This view is what drives kitchen-partner onboarding priority, rider fleet allocation across zones, and new-dark-kitchen siting decisions (a consistently high-demand zone is a candidate for additional kitchen-partner capacity to protect delivery time as order density grows).

In this analysis, total orders are compared across the 5 delivery zones to identify areas with the highest demand.

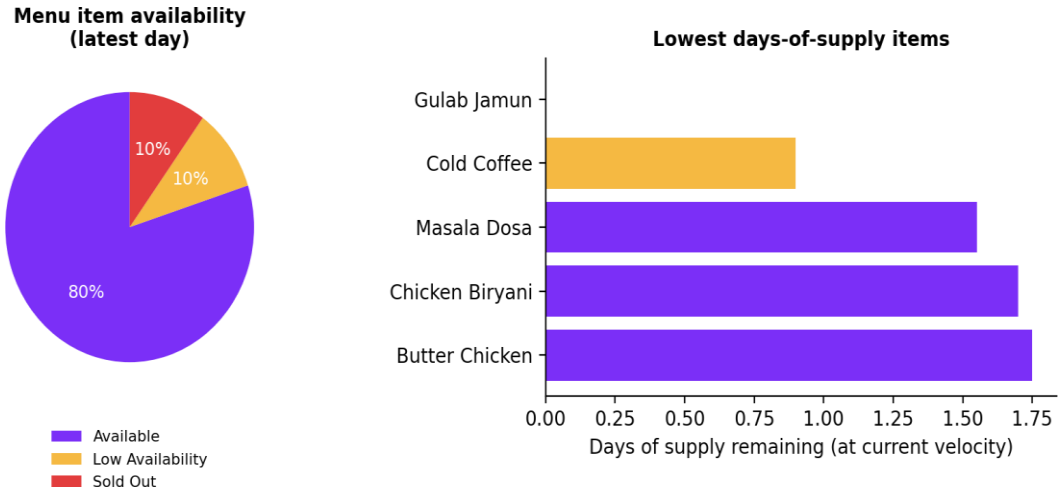
Andheri West - Mumbai recorded the highest demand with 1,772 orders, accounting for 37.7% of total orders. It was noticeably ahead of the other zones, showing that demand is concentrated more heavily in this particular zone.

To measure this concentration formally, the analysis uses the Herfindahl-Hirschman Index (HHI). HHI is calculated by squaring the order-share percentage of each zone and adding the results. A lower HHI indicates that demand is more evenly distributed, while a higher HHI indicates that demand is concentrated in a smaller number of zones.

The high demand concentration in the leading zone means that any capacity problem, rider shortage, or kitchen-partner disruption in this zone could have a significant impact on overall order fulfillment. Therefore, rider allocation, kitchen-partner onboarding, and backup fulfillment capacity should consider the level of demand in each zone.

With Andheri West - Mumbai carrying roughly a third of demand, a capacity disruption there (a kitchen-partner outage, a rider shortage, a local weather event) would affect total order volume far more than an equivalent disruption in the smallest zone — backup-rider pools and overflow fulfillment plans should be sized to that concentration, not spread evenly across all 5 zones.

# Q5 — Menu Item Availability & Stockout Risk




*Generated with pandas (aggregation) + matplotlib (visualization) directly from swiggy_order_data.csv.*

- Of 10 menu items, 8 are Available, 1 is in the Low Availability band, and 1 is fully Sold Out as of the latest day.
- The Sold Out item (Gulab Jamun) is a slower mover, which limits the order-volume impact — the same event on the #1 best-seller from Section 4 would be a materially bigger problem for both revenue and the habitual-reordering use case that keeps customers on the app.
- "Days of supply" (current prep-stock ÷ average daily orders) is a more actionable cut than a static availability flag: it converts kitchen readiness into a time-to-unavailability estimate per item, which is what actually triggers a kitchen-partner restock run.

The analysis found that out of 10 menu items, 8 were Available, 1 was in Low Availability, and 1 was Sold Out on the latest day. The Sold Out item was Gulab Jamun.

Days of Supply is more useful than looking only at the current availability flag because it converts kitchen readiness into an estimated time until unavailability. This helps identify which items need restocking sooner.

The analysis also uses the concept of a Reorder Point, which is the readiness level at which a new prep/restock cycle should be triggered. It is calculated using average daily demand, prep lead time, and safety buffer.

Overall, 2 out of 10 menu items were at Low Availability or below, showing that kitchen-readiness monitoring and timely restocking are important for preventing stockouts.

2 of 10 menu items are already at Low Availability or below in this short window — at real order volume, the same days-of-supply calculation run continuously per item per kitchen-partner is what would feed an automated restock queue, rather than a kitchen partner noticing a stockout after it has already cost sales and disappointed a customer expecting on-time delivery.

## Summary of Key Statistics

| Metric | Value (this window) |
|---|---|
| Average delivery time | 27.6 min (SLA target: 30 min) |
| Item-days exceeding SLA | 22.0% |
| Top single-window seller | Chicken Biryani (952 orders) |
| Top seller's share of total orders | 20.3% |
| Peak ordering window | Evening (5pm-9pm) (50.2% of orders) |
| Weekend vs. weekday order lift | +17.1% |
| Leading delivery zone | Andheri West - Mumbai (37.7% of demand) |
| Menu items at Low Availability or below | 2 of 10 |
| Highest SLA-risk zone | Hauz Khas - Delhi (25.2 → 31.9 min) |

# Next Steps: Moving From Descriptive → Predictive

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|---|---|
| **Delivery-time forecasting** | Predicts per-zone delivery time ahead of a shift, so rider allocation can be adjusted before an SLA breach happens, not after |
| **Demand forecasting** | Predicts per-item/per-zone order volume to set kitchen-partner reorder points automatically instead of reacting to a Low Availability alert |
| **Anomaly detection** | Flags a sudden delivery-time spike or unexpected demand surge at a specific kitchen-partner automatically, before a human notices the trend |

# Reference page

1. The Swiggy dataset –
   - Synthetic dataset from Claude AI.
2. Pandas documentation –
   - Data cleaning
   - Data manipulation and analysis.
3. Matplotlib documentation –
   - Creating graph
   - visualization.